

Curso:

1. Utilizar la técnica de mínimos cuadrados para ajustar los siguientes datos

x	-2	-1	0	1	2
y	-3	-1	0	3	5

mediante una función de la forma $y = ax + b$.

2. Determinar el conjunto de todos los $x \in \mathbb{R}^3$ tales que $A^{1961}x = [2 \ 0 \ 2]^T$, donde

$$a(-2) + b = -3$$

$$a(-1) + b = -1$$

$$a(0) + b = 0$$

$$a(1) + b = 3$$

$$a(2) + b = 5$$

$$-2a + b = -3$$

$$-a + b = -1$$

$$b = 0$$

$$a + b = 3$$

$$2a + b = 5$$

$$\begin{pmatrix} -2 & 1 \\ -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix}$$

$$\text{Aplico } A^T A \hat{x} = A^T b$$

$$\begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -3 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 10 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = \begin{pmatrix} 20 \\ 4 \end{pmatrix}$$

$$\begin{cases} 10\hat{a} = 20 \\ 5\hat{b} = 4 \end{cases}$$

$$\rightarrow \boxed{\hat{a} = 2}$$

$$\rightarrow \boxed{\hat{b} = \frac{4}{5}}$$

$$\boxed{y = 2x + \frac{4}{5}}$$

2. Determinar el conjunto de todos los $x \in \mathbb{R}^3$ tales que $A^{1961}x = \begin{bmatrix} 2 & 0 & 2 \end{bmatrix}^T$, donde

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

Nota

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

I) calculo los autovalores

$$p_A(\lambda) = \det \begin{pmatrix} 1-\lambda & 1 & 0 \\ 1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda)^2(2-\lambda) - [(1-\lambda)(1-\lambda)]$$

$$= (1-\lambda)(2-\lambda)(1-\lambda) - [0 + (1-\lambda) + (1-\lambda)]$$

$$= (1-\lambda)^2(2-\lambda) - 2(1-\lambda)$$

$$= (1-\lambda)[(1-\lambda)(2-\lambda) - 2]$$

$$= (1-\lambda)[2-\lambda-2\lambda+\lambda^2-2]$$

$$= (1-\lambda)[\lambda^2-2\lambda-\lambda]$$

$$= (1-\lambda)[\lambda^2-3\lambda]$$

$$= (1-\lambda)\lambda(\lambda-3)$$

$$\boxed{\lambda_1 = 1}$$

$$\boxed{\lambda_2 = 0}$$

$$\boxed{\lambda_3 = 3}$$

Busco los autoespacios de cada autovalor

$$S_1 = \text{Nul} \left(\begin{pmatrix} 1-1 & 1 & 0 \\ 1 & 2-1 & -1 \\ 0 & -1 & 1-1 \end{pmatrix} \right) \rightsquigarrow \text{Nul} \left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 0 \end{pmatrix} \right)$$

$$\text{Nul} \left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \right)$$

$$x_2 = 0$$

$$x_1 + x_2 - x_3 = 0$$

$$x_1 - x_3 = 0 \Rightarrow x_1 = x_3$$

$$\begin{aligned} [x_1 \ x_2 \ x_3]^T &= [x_1 \ 0 \ x_1]^T \\ &= x_1 [1 \ 0 \ 1]^T \end{aligned}$$

$$\boxed{\$1 = \text{gen} \{ [1 \ 0 \ 1]^T \}}$$

$$\text{Busco } \$0 = \text{Nul} \left(\begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \right) \sim \text{Nul} \left(\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \right)$$

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 - x_3 = 0 \end{cases} \rightarrow \boxed{x_2 = -x_1} \\ -x_1 - x_3 = 0 \Rightarrow \boxed{x_3 = -x_1}$$

$$\begin{aligned} [x_1 \ x_2 \ x_3]^T &= [x_1 \ -x_1 \ -x_1]^T \\ &= x_1 [1 \ -1 \ -1]^T \end{aligned}$$

$$\boxed{\$0 = \text{gen} \{ [1 \ -1 \ -1]^T \}}$$

$$\begin{aligned} \text{Busco } \$3 &= \text{Nul} \left(\begin{pmatrix} 1-3 & 1 & 0 \\ 1 & 2-3 & -1 \\ 0 & -1 & 1-3 \end{pmatrix} \right) = \text{Nul} \left(\begin{pmatrix} -2 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -2 \end{pmatrix} \right) \\ &= \text{Nul} \left(\begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \right) \end{aligned}$$

$$\begin{cases} -2x_1 + x_2 = 0 \rightarrow x_2 = 2x_1 \\ -x_2 - 2x_3 = 0 \rightarrow -2x_1 - 2x_3 = 0 \\ 2x_3 = -2x_1 \\ \boxed{x_3 = -x_1} \end{cases}$$

$$[x_1 \ x_2 \ x_3] = [x_1 \ 2x_1 \ -x_1] = x_1 [1 \ 2 \ -1]$$

$$\boxed{\$3 = \text{gen} \{ [1 \ 2 \ -1]^T \}}$$

$$B = \left\{ \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T \begin{bmatrix} 1 & -1 & -1 \end{bmatrix}^T \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}^T \right\}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & -1 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix}$$

$$D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix}^{-1}$$

Para calcular $A^{1961} x = \begin{bmatrix} 2 & 0 & 2 \end{bmatrix}^T$

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

Aplico

$$A^{1961} x = \alpha_1 A^{1961} v_1 + \alpha_2 A^{1961} v_2 + \alpha_3 A^{1961} v_3$$

$$A^{1961} x = \alpha_1 (0)^{1961} v_1 + \alpha_2 (1)^{1961} v_2 + \alpha_3 (3)^{1961} v_3$$

$$A^{1961} x = \alpha_2 v_2 + \alpha_3 (3)^{1961} v_3$$

Sabemos que $A^{1961} x = \begin{bmatrix} 2 & 0 & 2 \end{bmatrix}^T$

$$\alpha_2 v_2 + \alpha_3 (3)^{1961} v_3 = 0 v_1 + 2 v_2 + 0 v_3$$

$$\alpha_1 \in \mathbb{R}$$

$$\alpha_2 = 2$$

$$\alpha_3 = 0$$

La solución entonces es

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

$$x = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$x = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$x = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} \quad \alpha_1 \in \mathbb{R}$$

3. Hallar una matriz simétrica $A \in \mathbb{R}^{3 \times 3}$ tal que $[1 \ 0 \ 1]^T$ y $[1 \ 1 \ 1]^T$ son autovectores de A , $\text{traza}(A) = 7$ y $\det(A) = 12$.

4. Sea $A \in \mathbb{R}^{2 \times 3}$ la matriz definida por

Sabemos que $[1 \ 0 \ 1]^T$ y $[1 \ 1 \ 1]^T$ son autovectores de A

$$\text{Tra}(A) = 7 \quad \text{y} \quad \det(A) = 12$$

$$\langle [1 \ 0 \ 1] [1 \ 1 \ 1] \rangle = 1 + 0 + 1 = 2 \quad \therefore v_1 \text{ y } v_2 \text{ tienen asociado el mismo autovector}$$

$$\text{Construimos } v_3 = v_1 \times v_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix} = [0 - 1 \ 1 - 1]^T = [-1 \ 0 \ 1]^T$$

Sabemos que existe un $\lambda \in G(A)$ tal que $m(\lambda) = 2$

$\mu \in G(A)$ tal que $m(\mu) = 1$

$$\text{Tra}(A) = 7$$

y

$$\det(A) = 12$$

$$\lambda + \lambda + \mu = 7$$

$$\lambda \cdot \lambda \cdot \mu = 12$$

$$2\lambda + \mu = 7$$

$$\lambda^2 \cdot \mu = 12$$

$$\mu = 7 - 2\lambda$$

$$\lambda^2 (7 - 2\lambda) = 12$$

$$7\lambda^2 - 2\lambda^3 = 12$$

$$2\lambda^3 - 7\lambda^2 + 12 = 0$$

$$\text{aplicando } p(z) = 0$$

$\lambda = 2$ cumple con la Ecuación

$$\mu = 7 - 2(2) = 7 - 4 = 3$$

entonces $\lambda = 2$ con $m(2) = 2$ y $\mu = 3$ con $m(3) = 1$

Tenemos entonces

$$P = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Entonces

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{(-1)}$$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1/2 & -1 & 1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & 1/2 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & 0 \\ -3/2 & 0 & 3/2 \end{pmatrix}$$

$$A = \begin{pmatrix} 5/2 & 0 & -1/2 \\ 0 & 2 & 0 \\ -1/2 & 0 & 5/2 \end{pmatrix}$$